THE ACTION OF A DERIVATION ON A THICKENING EXISTS AND IS UNIQUE FOR ALGEBRAIC SPACES OVER A BASE SPACE

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 3747-3762 of sources/stacks-spaces-more-morphisms/pinned-original-source.tex; proof span is lines 3764-3779, the complete author proof (14 body lines).

Editorial note. Navigation stubs: 3 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. First order thickenings over a base algebraic space, conormal sheaves with their square-zero structure, and differentials of the morphisms; the construction is retained.

Source reference. Original lemma equation-D: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 3738–3740. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma lemma-difference-derivation: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 3689–3700. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma lemma-first-order-thickening-maps: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 1601–1621. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let S be a scheme. Let B be an algebraic space over S. Let $(a, a') : (X \subset X') \to (Y \subset Y')$ be a morphism of first order thickenings over B. Let*

$$\theta : a^*\Omega_{Y/B} \to \mathcal{C}_{X/X'}$$

be an $\mathcal{O}_X$-linear map. Then there exists a unique morphism of pairs $(b, b') : (X \subset X') \to (Y \subset Y')$ such that (1) and (2) of Lemma lemma-difference-derivation hold and the derivation D and θ are related by Equation (equation-D).

Date: stacks-project:spaces-more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Proof. Consider the map

$$\alpha = (a')^\sharp + D : a^{-1}\mathcal{O}_{Y'} \to \mathcal{O}_{X'}$$

where D is as in Equation (equation-D). As D is an $\mathcal{O}_B$-derivation it follows that α is a map of sheaves of $\mathcal{O}_B$-algebras. By construction we have $i_X^\sharp \circ \alpha = a^\sharp \circ i_Y^\sharp$ where $i_X : X \to X'$ and $i_Y : Y \to Y'$ are the given closed immersions. By Lemma lemma-first-order-thickening-maps we obtain a unique morphism $(a, b') : (X \subset X') \to (Y \subset Y')$ of thickenings over B such that $\alpha = (b')^\sharp$. Setting $b = a$ we win. $\square$